Ps 01

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1 Problem 1.22 DONE

1.1 Task

(Change of basis) Let $h, k \in \mathbb{R} \setminus \{0\}$, and bases $\mathcal{U}$ and $\mathcal{V}$ of $\mathbb{R}^3$ given by

$$\mathcal{U} = \left\{ \begin{pmatrix} 1 \\ h \\ k \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}, \text{ and } \mathcal{V} = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} h \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} k \\ 0 \\ 1 \end{pmatrix} \right\}.$$

Determine the set $\mathbb{F}$ of all the vectors in $\mathbb{R}^3$ that has the same coordinates in both $\mathcal{U}$ and $\mathcal{V}$. Is $\mathbb{F}$ a vector subspace of $\mathbb{R}^3$? (**Indication:** try and define a linear mapping such that its kernel is the set of vectors that we are looking for.)

1.2 Solution

Let $\alpha = (\alpha_1, \alpha_2, \alpha_3)$ be the shared coordinate vector. The condition "same coordinates in $\mathcal{U}$ and $\mathcal{V}$" means:

$$\begin{aligned} \alpha_1 u_1 + \alpha_2 u_2 + \alpha_3 u_3 &= \alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 \\ \alpha_1(u_1 - v_1) + \alpha_2(u_2 - v_2) + \alpha_3(u_3 - v_3) &= 0 \end{aligned}$$

Define $T : \mathbb{R}^3 \to \mathbb{R}^3$ by $T(\alpha) = M\alpha$, where M has columns $(u_i - v_i)$. Then $\mathbb{F} = \ker(T)$.

$$\begin{aligned} M &= \begin{bmatrix} 0 & -h & -k \\ h & 0 & 0 \\ k & 0 & 0 \end{bmatrix} \\ &\xrightarrow{R_3 \leftarrow R_3 - \frac{k}{h} R_2} \begin{bmatrix} 0 & -h & -k \\ h & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

From row 2: $h\alpha_1 = 0 \implies \alpha_1 = 0$ (since $h \neq 0$).

From row 1: $-h\alpha_2 - k\alpha_3 = 0 \implies \alpha_2 = -\frac{k}{h}\alpha_3$.

Let $\alpha_3 = t$, hence:

$$\mathbb{F} = \left\{ t \begin{pmatrix} 0 \\ -\frac{k}{h} \\ 1 \end{pmatrix} : t \in \mathbb{R} \right\}$$

$\mathbb{F}$ is a vector subspace of $\mathbb{R}^3$, as it is the kernel of the linear map $T(\alpha) = M\alpha$ (kernels of linear maps are subspaces by Proposition III.2.11).

2 Problem 1.31 DONE

2.1 Task

(Sequences as a vector space) Let $\mathbb{E} = \mathcal{F}(\mathbb{N}, \mathbb{R}) = \{(x_n)_{n=1}^{\infty} : x_n \in \mathbb{R},\ \forall\, n \in \mathbb{N}\}$ be the set of all real sequences $(x_n)_{n=1}^{\infty}$ (as defined in Example 1.11).

(i) Prove that $\mathbb{E}$, when equipped with the following operations:

$(x_n)_{n=1}^{\infty} + (y_n)_{n=1}^{\infty} := (x_n + y_n)_{n=1}^{\infty}$, $\lambda(x_n)_{n=1}^{\infty} := (\lambda x_n)_{n=1}^{\infty}$,

is a vector space and that $\dim \mathbb{E} = \infty$ (**Indication:** for the last part, it suffices to give a countable infinite set of linearly independent real sequences).

(ii) Prove that the set of all convergent sequences $\mathbb{F} = \mathcal{F}_c(\mathbb{N}, \mathbb{R}) := \{(x_n)_{n=1}^{\infty} \in \mathbb{E} : \lim_{n\to\infty} x_n \in \mathbb{R}\}$ is a subspace of $\mathbb{E}$. Does $\dim \mathbb{F} = \infty$?

(iii) Prove that the application

$\mathcal{L} : \mathbb{F} \to \mathbb{R}$

$(x_n)_n^{\infty} \mapsto \lim_{n\to\infty} x_n$

which maps convergent sequences to their limit, is a linear aplication from $\mathbb{E}$ to $\mathbb{R}$. Obtain as a corollary that the set $\mathbb{G} = \mathcal{F}_0(\mathbb{N}, \mathbb{R}) := \{(x_n)_{n=1}^{\infty} \in \mathbb{F} : \lim_{n\to\infty} x_n = 0\}$ of real sequences that converges to zero is a vector subspace of $\mathbb{F}$. Does $\dim \mathbb{G} = \infty$?

2.2 Solution

2.2.1 (i)

We verify the vector space axioms for $\mathbb{E} = \mathcal{F}(\mathbb{N}, \mathbb{R})$ over $\mathbb{R}$ with componentwise addition and scalar multiplication.

Let $(x_n), (y_n), (z_n) \in \mathbb{E}$ and $\lambda, \mu \in \mathbb{R}$.

1. Associativity of addition

   $((x_n)+(y_n))+(z_n) = (x_n+y_n)+(z_n) = ((x_n+y_n)+z_n) = (x_n+(y_n+z_n)) = (x_n)+((y_n)+(z_n))$

   Holds by associativity of addition in $\mathbb{R}$.

2. Commutativity of addition

   $(x_n) + (y_n) = (x_n + y_n) = (y_n + x_n) = (y_n) + (x_n)$

   Holds by commutativity of addition in $\mathbb{R}$.

3. Additive identity

   Let $0_{\mathbb{E}} = (0, 0, 0, \ldots)$ be the zero sequence. Then,

   $(x_n) + 0_{\mathbb{E}} = (x_n + 0) = (x_n)$

4. Additive inverse

   For each $(x_n) \in \mathbb{E}$, define $(-x_n) \in \mathbb{E}$. Then,

   $(x_n) + (-x_n) = (x_n + (-x_n)) = (0) = 0_{\mathbb{E}}$

5. Distributivity over scalar addition

   $(\lambda + \mu)(x_n) = ((\lambda + \mu)x_n) = (\lambda x_n + \mu x_n) = \lambda(x_n) + \mu(x_n)$

   Holds by distributivity in $\mathbb{R}$.

6. Distributivity over vector addition

   $\lambda((x_n) + (y_n)) = \lambda(x_n + y_n) = (\lambda x_n + \lambda y_n) = \lambda(x_n) + \lambda(y_n)$

   Holds by distributivity in $\mathbb{R}$.

7. Compatibility of scalar multiplication

   $\lambda(\mu(x_n)) = \lambda(\mu x_n) = (\lambda\mu x_n) = (\lambda\mu)(x_n)$

   Holds by associativity of multiplication in $\mathbb{R}$.

8. Scalar multiplicative identity

   $1 \cdot (x_n) = (1 \cdot x_n) = (x_n)$

All axioms are satisfied. Hence, $\mathbb{E}$ is a vector space over $\mathbb{R}$.

For $\dim \mathbb{E} = \infty$, we construct a countably infinite set of linearly independent sequences. For each $k \in \mathbb{N}$, define

$$e^{(k)} = (0, 0, \ldots, 0, \underbrace{1}_{k\text{-th position}}, 0, \ldots)$$

That is, the sequence whose n -th term is $e_n^{(k)} = \delta_{nk}$ (Kronecker delta).

Claim: for any finite $m \in \mathbb{N}$, the set $\{e^{(1)}, e^{(2)}, \ldots, e^{(m)}\}$ is linearly independent.

Suppose $\sum_{k=1}^{m} \alpha_k e^{(k)} = 0$ (the zero sequence). Evaluating at position j for any $1 \leq j \leq m$:

$$\sum_{k=1}^{m} \alpha_k e_j^{(k)} = \alpha_j = 0$$

Hence $\alpha_j = 0$ for all $1 \leq j \leq m$, so the set is linearly independent.

Since for every $m \in \mathbb{N}$ we can find m linearly independent vectors in $\mathbb{E}$, the dimension of $\mathbb{E}$ cannot be any finite number. Therefore, $\dim \mathbb{E} = \infty$.

2.2.2 (ii)

We verify the subspace criterion for $\mathbb{F} = \mathcal{F}_c(\mathbb{N}, \mathbb{R})$.

Zero vector. The zero sequence $(0, 0, 0, \ldots)$ satisfies $\lim_{n\to\infty} 0 = 0 \in \mathbb{R}$, hence $0 \in \mathbb{F}$.

Closure under addition. Let $(x_n)_{n=1}^{\infty}, (y_n)_{n=1}^{\infty} \in \mathbb{F}$. Then $\lim_{n\to\infty} x_n = a \in \mathbb{R}$ and $\lim_{n\to\infty} y_n = b \in \mathbb{R}$. By the sum rule for limits:

$$\lim_{n\to\infty} (x_n + y_n) = \lim_{n\to\infty} x_n + \lim_{n\to\infty} y_n = a + b \in \mathbb{R}$$

Hence $(x_n + y_n)_{n=1}^{\infty} \in \mathbb{F}$.

Closure under scalar multiplication. Let $(x_n)_{n=1}^{\infty} \in \mathbb{F}$ and $\lambda \in \mathbb{R}$. Then $\lim_{n\to\infty} x_n = a \in \mathbb{R}$. By the scalar rule for limits:

$$\lim_{n\to\infty} \lambda x_n = \lambda \lim_{n\to\infty} x_n = \lambda a \in \mathbb{R}$$

Hence $(\lambda x_n)_{n=1}^{\infty} \in \mathbb{F}$.

Therefore $\mathbb{F}$ is a subspace of $\mathbb{E}$.

Dimension. Consider the sequences $e^{(k)} = (0, \ldots, 0, 1, 0, \ldots)$ with 1 in position k, for each $k \in \mathbb{N}$. Each $e^{(k)}$ converges to 0, hence $e^{(k)} \in \mathbb{F}$. Suppose $\sum_{k=1}^{m} c_k e^{(k)} = 0$ for some finite collection. Then $c_k = 0$ for all k (as the k -th component of the sum equals c_k). Hence $\{e^{(k)}\}_{k\in\mathbb{N}}$ is an infinite linearly independent subset of $\mathbb{F}$, and therefore $\dim \mathbb{F} = \infty$.

2.2.3 (iii)

Claim: $\mathcal{L} : \mathbb{F} \to \mathbb{R}$ defined by $\mathcal{L}((x_n)_{n=1}^{\infty}) = \lim_{n\to\infty} x_n$ is a linear map.

Proof.

Let $(x_n), (y_n) \in \mathbb{F}$ and $\lambda \in \mathbb{R}$.

$$\begin{aligned}
\mathcal{L}(\lambda(x_n) + (y_n)) &= \mathcal{L}((\lambda x_n + y_n)) \\
&= \lim_{n\to\infty} (\lambda x_n + y_n) \\
&= \lambda \lim_{n\to\infty} x_n + \lim_{n\to\infty} y_n \\
&= \lambda \mathcal{L}((x_n)) + \mathcal{L}((y_n))
\end{aligned}$$

where the third equality uses the standard limit laws (linearity of the limit for convergent sequences). Hence $\mathcal{L}$ is linear.

Corollary: $\mathbb{G} = \mathcal{F}_0(\mathbb{N}, \mathbb{R})$ is a vector subspace of $\mathbb{F}$.

Observe that

$$\mathbb{G} = \left\{(x_n)_{n=1}^{\infty} \in \mathbb{F} : \lim_{n\to\infty} x_n = 0\right\} = \{(x_n) \in \mathbb{F} : \mathcal{L}((x_n)) = 0\} = \ker(\mathcal{L})$$

Since $\mathcal{L}$ is a linear map, $\ker(\mathcal{L})$ is a subspace of $\mathbb{F}$ (Proposition III.2.11(b)). Hence $\mathbb{G}$ is a vector subspace of $\mathbb{F}$.

Dimension: $\dim \mathbb{G} = \infty$.

Consider the sequences $e^{(k)} = (\delta_{n,k})_{n=1}^{\infty}$ for $k \in \mathbb{N}$, where $\delta_{n,k} = 1$ if $n = k$ and 0 otherwise. Each $e^{(k)}$ converges to 0, so $e^{(k)} \in \mathbb{G}$. For any finite subset $\{e^{(1)}, \ldots, e^{(m)}\}$, suppose

$$c_1 e^{(1)} + \cdots + c_m e^{(m)} = 0$$

Evaluating at position k gives $c_k = 0$ for each $k = 1, \ldots, m$. Hence any finite subset is linearly independent. Since $\mathbb{G}$ contains arbitrarily large finite linearly independent sets, $\dim \mathbb{G} = \infty$.

3 Problem 1.40 DONE

3.1 Task

(Matrix representation of linear applications and vector elements) Let $\mathbb{E} := \mathbb{R}^n$, and $\mathcal{E} = \{e_i\}_{i=1}^n \subset \mathbb{E}$ be the standard basis of $\mathbb{E}$.

(i) Let $\mathbb{F} := \mathbb{R}_{n-1}[x]$ be the vector space of real polynomials p of degree $\deg p \leq n-1$, construct a representation of $\mathbb{F}$ in $\mathbb{E}$. (**Indication:** assume that we have proven that $\{1, x, x^2, x^3, ..., x^{n-1}\} \subset \mathbb{F}$ forms a basis).

(ii) Let $D : \mathbb{F} \to \mathbb{F}$ be the *derivative mapping*: $D(p) = p'$ (where p' stands for the polynomial derivative of p). Prove that D is an endomorphism on $\mathbb{F}$ (i.e. $D \in \mathcal{L}(\mathbb{F})$) and give the corresponding matrix representation $[D]_{\mathcal{E}}^{\mathcal{E}}$. What is $\ker D$?

(iii) We now define the following application:

$L : \mathbb{F} \to \mathbb{F}$

$p \mapsto p' + \alpha p$

where $\alpha \in \mathbb{R} \setminus \{0\}$. Prove that $L \in \mathcal{L}(\mathbb{E})$, give $[L]_{\mathcal{E}}^{\mathcal{E}}$.

(iv) Let $q(x) = 5x^3 + 4x^2 + 5$, find $p(x) \in \mathbb{R}_3$ such that $p'(x) + 6p(x) = q(x)$.

(v) Can we conclude from (iii) the following: given $\alpha \in \mathbb{R} \setminus \{0\}$, for each $q \in \mathbb{R}_{n-1}[x]$ there exists an unique $p \in \mathbb{R}_{n-1}[x]$ such that $p' + \alpha p = q$?

3.2 Solution

3.2.1 Part (i)

The space $\mathbb{F} = \mathbb{R}_{n-1}[x]$ has basis $\{1, x, x^2, \ldots, x^{n-1}\}$, which has n elements. Hence $\dim \mathbb{F} = n = \dim \mathbb{R}^n$.

We construct a coordinate isomorphism $\varphi : \mathbb{F} \to \mathbb{E}$ by mapping each polynomial to its coefficient vector:

$$\varphi(a_0 + a_1 x + a_2 x^2 + \cdots + a_{n-1} x^{n-1}) = \begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_{n-1} \end{pmatrix}$$

This identifies each basis element x^k with the standard basis vector e_{k+1}:

$$\varphi(1) = e_1, \quad \varphi(x) = e_2, \quad \ldots, \quad \varphi(x^{n-1}) = e_n$$

3.2.2 Part (ii)

Claim: $D : \mathbb{F} \to \mathbb{F}$ defined by $D(p) = p'$ is an endomorphism on $\mathbb{F}$.

Proof of linearity. Let $p, q \in \mathbb{F}$ and $\lambda \in \mathbb{R}$.

$$D(\lambda p + q) = (\lambda p + q)' = \lambda p' + q' = \lambda D(p) + D(q)$$

Hence D is linear.

Proof that D maps $\mathbb{F}$ to $\mathbb{F}$. If $\deg p \leq n - 1$, then $\deg p' \leq n - 2 \leq n - 1$, so $p' \in \mathbb{F}$. Hence $D \in \mathcal{L}(\mathbb{F})$.

Matrix representation. We compute D on each basis element:

$$
\begin{aligned}
D(1) &= 0 \\
D(x) &= 1 \\
D(x^2) &= 2x \\
D(x^3) &= 3x^2 \\
&\vdots \\
D(x^{n-1}) &= (n-1)x^{n-2}
\end{aligned}
$$

Passing through the coordinate isomorphism φ, the k -th column of $[D]_{\mathcal{E}}^{\mathcal{E}}$ is the coordinate vector of $D(x^{k-1})$. Hence:

$$
[D]_{\mathcal{E}}^{\mathcal{E}} = \begin{bmatrix} 0 & 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 2 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 3 & \cdots & 0 \\ \vdots & & & & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & n-1 \\ 0 & 0 & 0 & 0 & \cdots & 0 \end{bmatrix}
$$

Kernel. $D(p) = 0$ means $p' = 0$, which holds if and only if p is a constant polynomial. Hence:

$$
\ker D = \{a \cdot 1 : a \in \mathbb{R}\} = \operatorname{span}\{1\}
$$

3.2.3 Part (iii)

Claim: $L : \mathbb{F} \to \mathbb{F}$ defined by $L(p) = p' + \alpha p$ is linear.

Proof. Let $p, q \in \mathbb{F}$ and $\lambda \in \mathbb{R}$.

$$
\begin{aligned}
L(\lambda p + q) &= (\lambda p + q)' + \alpha(\lambda p + q) \\
&= \lambda p' + q' + \alpha\lambda p + \alpha q \\
&= \lambda(p' + \alpha p) + (q' + \alpha q) \\
&= \lambda L(p) + L(q)
\end{aligned}
$$

If $\deg p \leq n-1$, then $\deg(p' + \alpha p) \leq n-1$, so L maps $\mathbb{F}$ to $\mathbb{F}$. Hence $L \in \mathcal{L}(\mathbb{F})$.

Matrix representation. Since $L = D + \alpha\,\mathrm{Id}$, we have $[L]_{\mathcal{E}}^{\mathcal{E}} = [D]_{\mathcal{E}}^{\mathcal{E}} + \alpha I_n$:

$$[L]_{\mathcal{E}}^{\mathcal{E}} = \begin{bmatrix} \alpha & 1 & 0 & 0 & \cdots & 0 \\ 0 & \alpha & 2 & 0 & \cdots & 0 \\ 0 & 0 & \alpha & 3 & \cdots & 0 \\ \vdots & & & & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & \alpha \end{bmatrix}$$

3.2.4 Part (iv)

We need $p(x) \in \mathbb{R}_3$ such that $p'(x) + 6p(x) = q(x) = 5x^3 + 4x^2 + 5$.

Here $n = 4$, $\alpha = 6$. Using the matrix from (iii), the problem becomes $[L]_{\mathcal{E}}^{\mathcal{E}} \cdot c = b$, where $b = \varphi(q) = (5, 0, 4, 5)^T$ and $c = (c_0, c_1, c_2, c_3)^T$ is the coordinate vector of p.

$$\begin{bmatrix} 6 & 1 & 0 & 0 \\ 0 & 6 & 2 & 0 \\ 0 & 0 & 6 & 3 \\ 0 & 0 & 0 & 6 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \\ 4 \\ 5 \end{bmatrix}$$

Solving by back substitution:

$$\begin{aligned} 6c_3 &= 5 & &\implies c_3 = \frac{5}{6} \\ 6c_2 + 3c_3 &= 4 & &\implies c_2 = \frac{4 - \frac{5}{2}}{6} = \frac{1}{4} \\ 6c_1 + 2c_2 &= 0 & &\implies c_1 = \frac{-\frac{1}{2}}{6} = -\frac{1}{12} \\ 6c_0 + c_1 &= 5 & &\implies c_0 = \frac{5 + \frac{1}{12}}{6} = \frac{61}{72} \end{aligned}$$

Therefore:

$$p(x) = \frac{61}{72} - \frac{1}{12}x + \frac{1}{4}x^2 + \frac{5}{6}x^3$$

3.2.5 Part (v)

Yes. The matrix $[L]_{\mathcal{E}}^{\mathcal{E}}$ is upper triangular with every diagonal entry equal to $\alpha \neq 0$. Therefore $\det([L]_{\mathcal{E}}^{\mathcal{E}}) = \alpha^n \neq 0$, so the matrix is invertible.

This means L is bijective: for every $q \in \mathbb{R}_{n-1}[x]$, the system $[L]_{\mathcal{E}}^{\mathcal{E}} \cdot c = \varphi(q)$ has a unique solution c, giving a unique $p = \varphi^{-1}(c) \in \mathbb{R}_{n-1}[x]$ with $p' + \alpha p = q$.

Equivalently, $\ker(L) = \{0\}$ (since the matrix is invertible), so L is injective. By the rank-nullity theorem, $\dim(\mathrm{im}(L)) = \dim(\mathbb{F}) - \dim(\ker(L)) = n - 0 = n$, hence L is surjective. Therefore L is bijective and the solution is unique.

4 Problem 1.42 DONE

4.1 Task

> (Dual space and dual basis in $\mathbb{R}^n$) In $\mathbb{E} = \mathbb{R}^n$ we are given the basis $\mathcal{V} = \{v_1, v_2, ..., v_n\}$, where $v_j = \sum_{k=1}^{j} k e_k$. Express the dual basis $\mathcal{V}^*$ in terms of the vectors e_j^* in the dual space $\mathbb{E}^*$.

4.2 Solution

The basis vectors are $v_j = \sum_{k=1}^{j} k\, e_k$, giving the coordinate matrix:

$$A = \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 0 & 2 & 2 & \cdots & 2 \\ 0 & 0 & 3 & \cdots & 3 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & n \end{bmatrix}$$

where $A_{kj} = k$ if $k \leq j$ and 0 otherwise.

We seek $v_i^* = \sum_{j=1}^{n} c_{ij} e_j^*$ such that $v_i^*(v_j) = \delta_{ij}$. This gives $CA = I$, hence $C = A^{-1}$.

Since A is upper triangular with diagonal entries $1, 2, \ldots, n$, we find A^{-1} by back substitution. Denote $C = (c_{ij})$. From $CA = I$, the \$i\$-th row of C satisfies:

$$\sum_k c_{ik} A_{kj} = \delta_{ij}$$

For diagonal entries: $c_{ii} \cdot i = 1$, so $c_{ii} = \frac{1}{i}$.

For $j > i$: $c_{ij} \cdot j + c_{i,j-1} \cdot (j-1) + \cdots = 0$ (only terms with $k \leq j$ contribute, and $c_{ik} = 0$ for $k < i$).

Working this out, for $j = i + 1$:

$$c_{i,i+1} \cdot (i+1) + c_{ii} \cdot i = 0 \implies c_{i,i+1} = -\frac{c_{ii} \cdot i}{i+1} = -\frac{1}{i+1}$$

For $j = i + 2$:

$$c_{i,i+2} \cdot (i+2) + c_{i,i+1} \cdot (i+1) + c_{ii} \cdot i = 0 \implies c_{i,i+2} \cdot (i+2) + (-1) + 1 = 0 \implies c_{i,i+2} = 0$$

Continuing, all further entries vanish: $c_{ij} = 0$ for $j \geq i+2$.

Therefore:

$$C = A^{-1} = \begin{bmatrix} 1 & -\frac{1}{2} & 0 & 0 & \cdots & 0 \\ 0 & \frac{1}{2} & -\frac{1}{3} & 0 & \cdots & 0 \\ 0 & 0 & \frac{1}{3} & -\frac{1}{4} & \cdots & 0 \\ \vdots & & & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \frac{1}{n-1} & -\frac{1}{n} \\ 0 & 0 & 0 & \cdots & 0 & \frac{1}{n} \end{bmatrix}$$

Reading off the rows, the dual basis vectors are:

$$\begin{aligned} v_1^* &= e_1^* - \frac{1}{2} e_2^* \\ v_2^* &= \frac{1}{2} e_2^* - \frac{1}{3} e_3^* \\ v_3^* &= \frac{1}{3} e_3^* - \frac{1}{4} e_4^* \\ &\vdots \\ v_{n-1}^* &= \frac{1}{n-1} e_{n-1}^* - \frac{1}{n} e_n^* \\ v_n^* &= \frac{1}{n} e_n^* \end{aligned}$$

In general: $v_i^* = \frac{1}{i} e_i^* - \frac{1}{i+1} e_{i+1}^*$ for $i = 1, \ldots, n-1$, and $v_n^* = \frac{1}{n} e_n^*$.

5 Problem 1.43 DONE

5.1 Task

(Dual space and dual basis in $\mathbb{R}_n[X]$)

Let $\mathbb{E} := \mathbb{R}_1[X]$ be the space of real polynomials $p(X)$ with $\deg p \leq 1$ (Suppose that we have already proven that the standard basis on $\mathbb{E}$, i.e. $\{1, X\}$, is indeed a basis). Consider functions $f, g, h : \mathbb{E} \to \mathbb{K}$ defined by

$f(p(X)) = p(0)$, $g(p(X)) = p(1)$, and $h(p(X)) = \int_0^1 p(X)\, dX$, $p(X) \in \mathbb{R}_1[X]$.

(i) Prove that $f, g, h \in \mathbb{E}^*$, and that f and g form a basis on $\mathbb{E}^*$.

(ii) Express h as a linear combination of f and g, is the linear combination unique?

(iii) Give the dual basis $\{e_1^*, e_2^*\}$ of the standard basis $\{1, X\}$ (**Indication:** try and obtain these vectors as a linear combination of the basis vectors $\{f, g\}$).

5.2 Solution

5.2.1 Part (i)

Claim: $f, g, h \in \mathbb{E}^*$.

We must show each map is linear, i.e. for $p, q \in \mathbb{E}$ and $\lambda \in \mathbb{R}$:

Linearity of f:

$$f(\lambda p + q) = (\lambda p + q)(0) = \lambda p(0) + q(0) = \lambda f(p) + f(q)$$

Linearity of g:

$$g(\lambda p + q) = (\lambda p + q)(1) = \lambda p(1) + q(1) = \lambda g(p) + g(q)$$

Linearity of h:

$$h(\lambda p + q) = \int_0^1 (\lambda p(X) + q(X))\, dX = \lambda \int_0^1 p(X)\, dX + \int_0^1 q(X)\, dX = \lambda h(p) + h(q)$$

Hence $f, g, h \in \mathbb{E}^*$.

Claim: f and g form a basis of $\mathbb{E}^*$.

Since $\dim \mathbb{E}^* = \dim \mathbb{E} = 2$, it suffices to show f and g are linearly independent. Suppose $\alpha f + \beta g = 0$ (the zero functional). Then for all $p \in \mathbb{E}$:

$$\alpha f(p) + \beta g(p) = 0$$

Evaluating at $p = 1$ (the constant polynomial):

$$\alpha \cdot 1 + \beta \cdot 1 = 0 \implies \alpha + \beta = 0$$

Evaluating at $p = X$:

$$\alpha \cdot 0 + \beta \cdot 1 = 0 \implies \beta = 0$$

Hence $\alpha = 0$ and $\beta = 0$. Therefore $\{f, g\}$ is linearly independent, and since $|\{f, g\}| = \dim \mathbb{E}^*$, it is a basis of $\mathbb{E}^*$.

5.2.2 Part (ii)

We seek $\alpha, \beta \in \mathbb{R}$ such that $h = \alpha f + \beta g$, meaning for all $p(X) = a + bX \in \mathbb{E}$:

$$h(p) = \alpha f(p) + \beta g(p)$$

Compute h, f, g on a general polynomial $p(X) = a + bX$:

$$\begin{aligned} f(p) &= p(0) = a \\ g(p) &= p(1) = a + b \\ h(p) &= \int_0^1 (a + bX)\, dX = a + \frac{b}{2} \end{aligned}$$

So we need:

$$a + \frac{b}{2} = \alpha a + \beta(a + b)$$

for all $a, b \in \mathbb{R}$. Comparing coefficients:

$$\begin{aligned} a: \quad & \alpha + \beta = 1 \\ b: \quad & \beta = \frac{1}{2} \end{aligned}$$

Hence $\beta = \frac{1}{2}$ and $\alpha = \frac{1}{2}$.

$$h = \frac{1}{2} f + \frac{1}{2} g$$

The linear combination is unique because $\{f, g\}$ is a basis of $\mathbb{E}^*$ (every vector has a unique representation in terms of a basis).

5.2.3 Part (iii)

The dual basis $\{e_1^*, e_2^*\}$ of the standard basis $\{1, X\}$ is defined by:

$$\begin{aligned} e_1^*(1) = 1, \quad e_1^*(X) = 0 \\ e_2^*(1) = 0, \quad e_2^*(X) = 1 \end{aligned}$$

In other words, e_1^* picks out the constant coefficient and e_2^* picks out the coefficient of X.

We express $e_1^* = \alpha f + \beta g$ and $e_2^* = \gamma f + \delta g$.

Finding e_1^*:

$$\begin{aligned} e_1^*(1) = \alpha f(1) + \beta g(1) = \alpha + \beta = 1 \\ e_1^*(X) = \alpha f(X) + \beta g(X) = \alpha \cdot 0 + \beta \cdot 1 = \beta = 0 \end{aligned}$$

Hence $\beta = 0$ and $\alpha = 1$:

$$e_1^* = f$$

Finding e_2^*:

$$\begin{aligned} e_2^*(1) = \gamma f(1) + \delta g(1) = \gamma + \delta = 0 \\ e_2^*(X) = \gamma f(X) + \delta g(X) = \gamma \cdot 0 + \delta \cdot 1 = \delta = 1 \end{aligned}$$

Hence $\delta = 1$ and $\gamma = -1$:

$$e_2^* = -f + g$$

Verification: For $p(X) = a + bX$:

$$\begin{aligned} e_1^*(p) = f(p) = p(0) = a \quad \checkmark \\ e_2^*(p) = -f(p) + g(p) = -a + (a + b) = b \quad \checkmark \end{aligned}$$